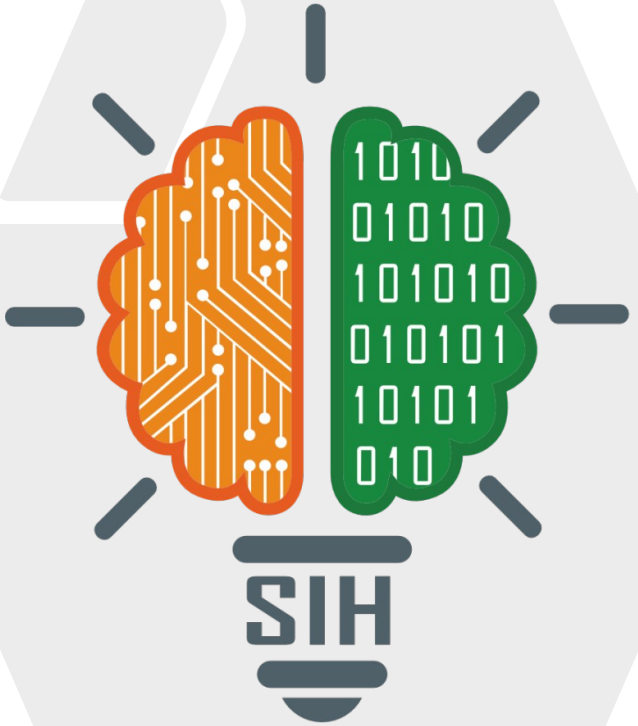
SMART INDIA HACKATHON 2026

SCOURGE

Ask the bank, not the paper.



- **Problem Statement ID –** SIH26265
- **Problem Statement Title –** Streamline of Security Deposit / FDR Portal
- **Theme –** Smart Automation
- **PS Category –** Software
- **Team ID –** 199322
- **Team Name (Registered on portal) –** Arnavax#

SCOURGE — ASK THE BANK, NOT THE PAPER

Proposed Solution (Describe your Idea/Solution/Prototype)

An institution proves what its bank's own server says about one FDR. AICTE sees only that record.

Detailed explanation of the proposed solution

AICTE sends a **single-use challenge** for one FDR. The institution logs into its own net banking inside one TLS session whose keys are **split with AICTE's verifier** (MPC-TLS). Password, OTP, PAN and other accounts never leave the institution.

How it addresses the problem

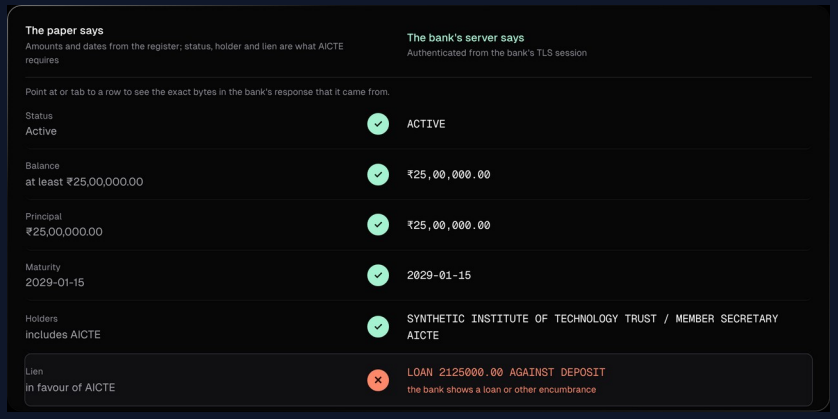
Paper is checked once and trusted for 10 years. Scourge **re-checks the bank any time and before every release**: a loan, early encashment, missing pledge (lien) or fake FDR shows up as a **Discrepancy**.

Innovation and uniqueness of the solution

Works on every bank's existing net-banking page, and API-ready. Three evidence tiers: bank-session proof, then bank-signed e-advice, then branch confirmation. Split key: the institution cannot edit what the bank said; fakes and replays are rejected.

LIVE PROTOTYPE · SYNTHETIC BANK

The bank's server contradicts the paper.



× HIDDEN LOAN CAUGHT

Every value traces to the bank's own bytes

The complete FDR lifecycle in one portal (all five stages run in the prototype)

1 FILE — Institution files online; renewals linked

→ **2 PROVE** — Bank-session proof on AICTE's challenge

→ **3 MONITOR** — T-90/60/30/7 alerts, escalation; Table 1.4 adequacy

→ **4 DECIDE** — Verdict + evidence; hash-chained audit

→ **5 PLEDGE RELEASE** — Dept clearances, 2 signers, holds; QR-signed letter

Why now: **₹201.42 crore of FDRs "do not appear in IDFC's records"** (CBI chargesheet, Oct 2026); AICTE joint FDRs found encashed or loaned against without consent.

SCOURGE

TECHNICAL APPROACH

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Technologies to be used

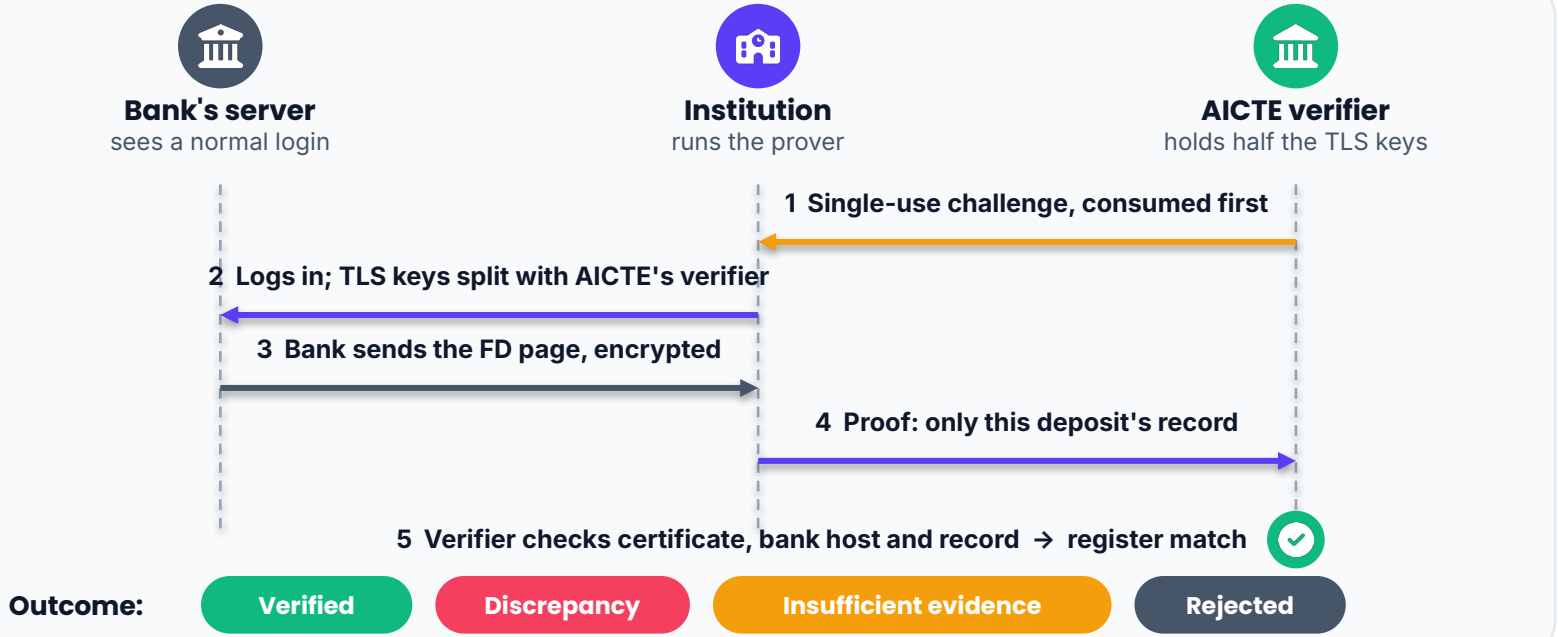
Proof layer
TLSNotary MPC-TLS · Rust
Selective disclosure · rustls

Backend
Python · FastAPI · SQLite
AES-GCM vault · hash-chained audit

Frontend
Server-rendered HTML · JS
Live 3-party proof view

Ops & QA
Docker · nginx · GitHub CI
173 tests · 2FA · audits

Methodology and process for implementation (Flow Charts/Images/ working prototype)



WATCH THE 90-SECOND DEMO



Click to play here, or scan / click the QR for YouTube
Real proofs on a synthetic bank

SCOURGE

FEASIBILITY AND VIABILITY

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Analysis of the feasibility of the idea

0.58 s
per bank proof
median of 60 runs · synthetic bank

13 / 16
real TLS sessions, 6 Indian banks
public pages, not FD pages

173 / 173
automated tests pass
incl. attack, forgery, 2FA, vault tests

₹0
software licences
one small server [estimate]

- **Technical:** Open-source TLSNotary (MIT/Apache-2.0); runs on one small server
- **Operational:** Institution uses its own login; no bank integration needed
- **Viability:** Adopted by an AICTE circular; covers the existing FDR book

Potential challenges and risks

Strategies for overcoming these challenges

Bank page hides liens, or bytes are withheld	Fallbacks: bank-signed e-advice, then branch confirmation; never a guess
Fake bank site or a replayed old proof	Rejected: bank certificate + single-use, 15-min challenge
Proof of a different FD, or an edited screen	Rejected / no effect: FD bound to challenge; proof covers bank bytes
Bank needs browser-bound session / TLS 1.3	Browser-extension prover; bank marked unsupported until then

ROADMAP 1 Live proof on a real bank's FD page → 2 Standard FDR-pledge schema for all banks → 3 Pilot with an AICTE regional office → 4 Bank confirmation API as banks onboard; e-BG for new deposits

SCOURGE

IMPACT AND BENEFITS

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Potential impact on the target audience

AICTE RB officers
A dated proof from the bank in seconds, not a letter in weeks

AICTE finance
The whole FDR book is re-checkable, before every release

Institutions
One short proof from their own net banking; Hindi or English

Students
The deposit that protects them is checked across all 10 years

Benefits of the solution (social, economic, environmental, etc.)

TODAY	WITH SCOURGE
Branch letter or uploaded screenshot	**The bank's own server, proven**
Checked once, trusted for 10 years	**Re-checked any time, before every release**
Weeks per confirmation	**Under a second per proof (synthetic bank)**
Hidden loans found at release, or never	**Pledge checked at the bank; two-officer release**

TESTED ON 180 FRESH DEPOSITS
Random amounts, dates, formats · real proofs · synthetic bank

119 / 119	bad deposits the upload portal approved
0 / 119	bad deposits Scourge verified
0 / 61	honest deposits Scourge accused

Social
students' deposits stay intact

Economic
stops loss before release

Environmental
no paper letters or couriers

Governance
2FA, encrypted vault, chained audit

SCOURGE

RESEARCH AND REFERENCES

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Details / Links of the reference and research work

The problem is real

CBI: ₹201.42 cr of FDRs not in IDFC's records (ThePrint, 7 Oct 2026)
theprint.in › cbi-says-haryana-pollution-board...

CBI on AICTE: joint FDRs encashed or loaned without consent
deccanherald.com › greater-transparency-aicte...

Regulation

AICTE guidelines 2024-27: cl. 6.2(g) 10-year release; cl. 11.8
indiankanoon.org/doc/173919027

Account Aggregator users must be regulated FIUs (DFS)
financialservices.gov.in › account-aggregator...

Technology

TLSNotary: MPC-TLS proofs of web data, MIT / Apache-2.0
tlsnotary.org

NeSL e-BG for security deposits: new instruments only (WDRA)
wdra.gov.in › ebg_requirement-27-08-2025

PROJECT RESOURCES

Demo video (90 s): youtu.be/8HoplhFOGCg
Prototype: Scourge v1.0.0-rc1 · 173 / 173 tests · synthetic bank
Code + evidence: github.com/Lakki007/Arnavax-Scourge-SIH26265

Scourge vs existing approaches

	Upload & approve	Bank letter	Account Aggregator	NeSL e-BG	Scourge
Uses the bank's own data	✗	✓	✓	✓	✓
No bank API or agreement	✓	✓	✗	✗	✓
Institution cannot forge it	✗	~	✓	✓	✓
Covers existing FDR book	✓	✓	✓	✗	✓
Re-check in seconds	✗	✗	✓	✓	✓
Shown on a real bank today	✓	✓	✓	✓	Next step

~ a bank letter can still be forged. A live real-bank proof is the next roadmap step (slide 4).